Physics Unit 2 · Practice Quiz 2

Static and kinetic friction · inclined planes · universal gravitation · circular motion · Newton's laws

Name: _____ Date: _____ Period: _____

Instructions. Part A: 20 multiple-choice questions, 2 points each (40 points). Part B: 4 free-response questions, 10 points each (40 points). Total 80 points. Suggested time: 50 minutes.

Reference. $g = 9.81\ \text{m/s}^2$ · $G = 6.67 \times 10^{-11}\ \text{N·m}^2/\text{kg}^2$ · $M_E = 5.97 \times 10^{24}\ \text{kg}$ · $R_E = 6.38 \times 10^{6}\ \text{m}$. Give answers to 3 significant figures with units. Show your work on the free-response questions. A calculator may be used (degree mode).

Part A · Multiple choice (2 points each)

1. A car speeds up on a level road without the tires skidding. The horizontal force that accelerates the car is

A the engine pushing on the car
B the normal force from the road
C static friction from the road on the tires, pointing forward
D kinetic friction from the road, pointing backward

2. A box weighs 250 N. A horizontal push must reach 110 N before the box starts to slide. What is μ_s?

A 0.045
B 2.3
C 0.44
D 4.3

3. You pull a 30 kg sled with a rope at 30° above the horizontal with a 100 N force. What is the normal force from the ground?

A 294 N
B 344 N
C 208 N
D 244 N

4. Two boxes, 5 kg and 20 kg, made of the same material, slide across the same floor with the same starting speed. Which slides farther before it stops?

A The 20 kg box
B They slide the same distance
C The 5 kg box
D The 20 kg box, because it has more momentum

5. A 118 N horizontal push keeps a 40 kg crate sliding at constant velocity. What is μ_k?

A 3.3
B 3.0
C 0, because the net force is zero
D 0.30

6. A car skids with locked wheels from 25 m/s on a level road with $\mu_k = 0.60$. How long does it take to stop?

A 2.55 s
B 1.53 s
C 53.1 s
D 4.25 s

7. You slowly tilt a board until a block on it just begins to slip, at angle θ. What is μ_s?

A tan θ
B sin θ
C cos θ
D θ in radians

8. A 12 kg box sits on a 25° incline. What is the component of its weight parallel to the incline?

A 118 N
B 107 N
C 49.8 N
D 54.9 N

9. μ_s between a block and a board is 0.60. The board is tilted slowly. At what angle does the block start to slide?

A 31.0°
B 53.1°
C 59.0°
D 36.9°

10. A 4.0 kg box rests on a 20° ramp ($\mu_s = 0.60$) and does not slide. What is the friction force on the box?

A 13.4 N up the slope
B 22.1 N up the slope
C 36.9 N up the slope
D 13.4 N down the slope

11. Astronauts in the International Space Station float because

A gravity at that height is almost zero
B there is no gravity in space
C they and the station are in free fall around Earth together
D the station's engines cancel gravity

12. Two objects attract each other with a gravitational force F. Both masses are doubled and the distance is doubled. What is the new force?

A 2F
B F
C 16F
D 4F

13. What is the gravitational field strength 400 km above Earth's surface? ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.38 \times 10^6$ m)

A 2,490 m/s²
B 9.78 m/s²
C 8.66 m/s²
D 9.23 m/s²

14. Two satellites orbit Earth at the same height. One has twice the mass of the other. Compared with the lighter one, the heavier satellite's orbital speed is

A √2 times as large
B the same
C twice as large
D half as large

15. A ball whirled in a horizontal circle on a string is released when the string breaks. It moves

A in a curve that continues around the circle
B in a spiral outward
C straight outward, away from the center
D in a straight line along the tangent at the point of release

16. If the speed of an object moving in a circle is doubled (same radius), the centripetal force needed becomes

A 4 times as large
B half as large
C unchanged
D 2 times as large

17. A 0.5 kg ball on a 0.8 m string is swung in a vertical circle. At the TOP its speed is 4.0 m/s. What is the tension in the string?

A 10.0 N **B** 5.09 N
C 14.9 N **D** 4.91 N

18. A 1000 kg car goes over the top of a hill (radius of curvature 40 m) at 12 m/s. What is the normal force on the car?

A 13,400 N **B** 6,210 N
C 9,810 N **D** 3,600 N

19. A 50 kg student stands on a scale in an elevator that is accelerating downward at 1.5 m/s². What does the scale read?

A 75.0 N **B** 566 N
C 490 N **D** 416 N

20. A truck and a small car collide head-on. During the collision, the force on the car is

A equal in size to the force on the truck
B larger than the force on the truck
C smaller than the force on the truck
D zero if the truck is moving faster

Part B · Free response (10 points each)

21. Pulling at an angle with friction (10 points)

A 30 kg sled is pulled across level snow by a rope that makes 30° with the horizontal. The tension is 150 N and $\mu_k = 0.20$.



(a) Draw and label the free-body diagram of the sled. **[2]**

(b) Find the normal force on the sled. **[2]**

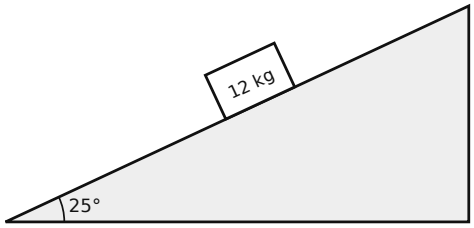
(c) Find the kinetic friction force. **[2]**

(d) Find the acceleration of the sled. **[2]**

(e) Explain why pulling up at an angle needs less force than pushing down at the same angle. **[2]**

22. A crate on a ramp (10 points)

A 12 kg crate is on a 25° ramp. $\mu_s = 0.40$ and $\mu_k = 0.25$.



(a) The crate is placed on the ramp at rest and let go. Does it stay at rest or slide? Show the test you used. **[2]**

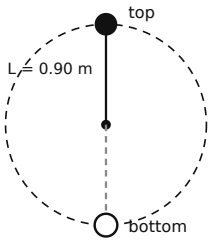
(b) Find its acceleration as it slides down. **[3]**

(c) What force parallel to the ramp pushes the crate UP the ramp at constant speed? **[3]**

(d) What is the smallest force parallel to the ramp (pointing up the slope) that keeps the crate at rest? **[2]**

23. Vertical circle (10 points)

A 0.40 kg ball on a 0.90 m string is swung in a vertical circle.



(a) Draw the free-body diagram of the ball at the top and at the bottom of the circle. **[2]**

(b) What is the minimum speed at the top for the string to stay tight? **[2]**

(c) Find the tension at the top when the ball moves at 4.0 m/s. **[2]**

(d) Find the tension at the bottom when the ball moves at 6.0 m/s. **[2]**

(e) Where is the string most likely to break? Explain. **[2]**

24. Universal gravitation (10 points)

Use $G = 6.67 \times 10^{-11}$ N·m²/kg².

(a) A 50 kg lead sphere and a 2.0 kg lead sphere are 0.30 m apart (center to center). Find the gravitational force between them. **[2]**

(b) Two objects attract with force F. Both masses are doubled and the distance between them is tripled. Express the new force in terms of F. **[2]**

(c) A planet has 5 times Earth's mass and 2 times Earth's radius. Find g at its surface and the weight of a 60 kg student there. **[3]**

(d) At what height above Earth's surface is g = 2.45 m/s²? **[3]**